GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Monocarboxylic Acid Transport (GO:001571	0.25316227	50	6.202e−10	3.331e−06	SLC5A12:15 SLC16A5:222 SLC10A3:282 SLC5A8:439 SLCO1A2:474 SLC16A9:568
Organic Hydroxy Compound Transport (GO:0	0.21918957	36	5.435e−06	1.460e−02	SLC10A3:282 SLC5A8:439 SLCO1A2:474 SLC22A1:690 ABCG5:834 SLCO1B1:860
Lysosome Localization (GO:0032418)	−0.23448608	27	2.500e−05	4.476e−02	SPAG9:18 GRP:659 NDEL1:743 KIT:807 HPS6:875 BORCS7:1032
Bile Acid And Bile Salt Transport (GO:00	0.28682788	16	7.157e−05	9.611e−02	SLC10A3:282 SLCO1A2:474 SLCO1B1:860 SLC51B:931 SLC51A:939 SLC10A6:1062
Organic Cation Transport (GO:0015695)	0.20083671	30	1.420e−04	1.525e−01	SLC22A16:24 SLC44A5:401 SLC6A12:596 SLC22A1:690 SLC18B1:994 RALBP1:1037
Positive Regulation Of Cell Population P	−0.05836496	328	3.177e−04	2.844e−01	MAPK15:35 GDNF:36 KMT2D:75 IL5RA:88 ZFPM2:99 IGF1R:101
tRNA Methylation (GO:0030488)	0.17542527	34	4.045e−04	3.103e−01	TARBP1:259 TRMT10A:592 TRMT9B:1242 LCMT2:1292 METTL2B:1489 TRMT10C:1800
Carboxylic Acid Transport (GO:0046942)	0.14291862	47	7.095e−04	4.234e−01	SLC5A12:15 SLC16A5:222 SLC36A2:227 SLC6A6:519 SLC36A1:564 SLC16A9:568
Small Nucleolar Ribonucleoprotein Comple	−0.32775028	9	6.632e−04	4.234e−01	TNIP1:139 RUVBL2:704 SHQ1:1577 RUVBL1:1640 PIH1D1:1854 ZNHIT3:2199
RNA Methylation (GO:0001510)	0.14016765	48	7.917e−04	4.252e−01	TARBP1:259 TRMT10A:592 FTSJ3:618 TRMT9B:1242 LCMT2:1292 CMTR1:1326
Energy Derivation By Oxidation Of Organi	−0.20015078	22	1.161e−03	4.372e−01	COX15:42 COX6A1:345 PPARGC1A:1141 COX4I1:1162 CYC1:1463 NDUFS4:1526
Mast Cell Activation Involved In Immune	−0.37570679	6	1.438e−03	4.372e−01	GRP:659 KIT:807 NR4A3:1537 PIK3CD:1596 FCER1A:2348 PIK3CG:2551
Mast Cell Degranulation (GO:0043303)	−0.37570679	6	1.438e−03	4.372e−01	GRP:659 KIT:807 NR4A3:1537 PIK3CD:1596 FCER1A:2348 PIK3CG:2551
Mast Cell Mediated Immunity (GO:0002448)	−0.37570679	6	1.438e−03	4.372e−01	GRP:659 KIT:807 NR4A3:1537 PIK3CD:1596 FCER1A:2348 PIK3CG:2551
Monoatomic Cation Homeostasis (GO:005508	0.11946207	61	1.273e−03	4.372e−01	CYBRD1:66 KCNH2:133 SLC12A8:160 CLDN16:186 LETM1:383 CNNM4:537
Neuron Differentiation (GO:0030182)	−0.08695278	114	1.387e−03	4.372e−01	RTN4:132 USH2A:265 SPINK5:358 TENM1:431 ERBB4:453 WNT7B:470
Peptidyl−Tyrosine Phosphorylation (GO:00	−0.12634054	57	9.858e−04	4.372e−01	IGF1R:101 ERBB4:453 ABI3:473 EFEMP1:593 DDR1:620 RELN:634
Ubiquitin−Dependent Protein Catabolic Pr	−0.06180434	226	1.465e−03	4.372e−01	AMBRA1:34 TBL1XR1:40 PSMA7:138 USP33:180 PSMD7:184 SPSB1:191
Positive Regulation Of T Cell Activation	−0.10627397	74	1.606e−03	4.539e−01	CD276:373 THY1:500 TNFSF11:669 CD274:721 ZP4:784 IL21:845
Stem Cell Differentiation (GO:0048863)	−0.17775037	26	1.716e−03	4.609e−01	LMBR1L:55 TP53:297 FOXO4:499 KIT:807 FGFR2:955 MEOX1:1014
Calcium−Independent Cell−Cell Adhesion V	0.26764411	11	2.118e−03	4.783e−01	CLDN16:186 CLDN8:211 CLDN18:460 CLDN20:1460 CLDN7:1466 CLDN23:2614
Intracellular Monoatomic Ion Homeostasis	0.20519185	19	1.967e−03	4.783e−01	CLDN16:186 LETM1:383 CNNM4:537 SLC4A1:670 RHAG:914 RHCG:1513
Lipid Transport (GO:0006869)	0.10030826	78	2.239e−03	4.783e−01	EHHADH:116 ABCB11:144 ABCA2:161 SLC10A3:282 SLCO1A2:474 MTTP:506
Positive Regulation Of Double−Strand Bre	0.28281818	10	1.959e−03	4.783e−01	KMT5C:924 WRAP53:967 SMCHD1:1003 KMT5B:1369 PRKDC:1509 SHLD2:1655
Regulation Of Synaptic Vesicle Exocytosi	−0.25404122	12	2.315e−03	4.783e−01	RAP1A:1 VPS18:77 RIMS1:481 SYN1:586 RIMS2:936 SNAPIN:1457
Ribosome Biogenesis (GO:0042254)	−0.08360945	112	2.306e−03	4.783e−01	TNIP1:139 PAK1IP1:195 NOB1:372 CINP:443 WDR75:463 BYSL:477
Positive Regulation Of Defense Response	0.33090261	7	2.433e−03	4.836e−01	PGC:5 EMILIN2:215 GRN:723 KLK5:1274 KLK7:1275 EMILIN1:5581
Positive Regulation Of Transcription By	−0.03595721	628	2.521e−03	4.836e−01	ESRRG:4 NR5A2:12 GDNF:36 ABHD14B:38 TBL1XR1:40 STING1:41
rRNA Metabolic Process (GO:0016072)	−0.11060048	62	2.637e−03	4.884e−01	NOB1:372 BYSL:477 NOP58:748 DDX54:984 XRN2:1125 ESF1:1134
Antigen Receptor−Mediated Signaling Path	−0.09003281	90	3.228e−03	4.917e−01	CD276:373 NFATC2:457 TRAF6:497 THY1:500 STOML2:581 EIF2B2:625
Cardiac Right Ventricle Morphogenesis (G	−0.42305169	4	3.387e−03	4.917e−01	ZFPM2:99 GATA4:909 JAG1:960 SOX4:1926 NA NA
Centromeric Sister Chromatid Cohesion (G	0.42507066	4	3.238e−03	4.917e−01	MEIKIN:319 SGO1:740 BUB1:1114 BUB1B:1623 NA NA
Ear Morphogenesis (GO:0042471)	−0.22680159	14	3.309e−03	4.917e−01	TIFAB:68 USH1C:79 FGFR2:955 OSR2:2716 EPHB2:2746 NOG:2781
Fat−Soluble Vitamin Metabolic Process (G	0.21310895	16	3.174e−03	4.917e−01	CYP11A1:82 CYP2R1:92 CYP1A1:318 LRAT:678 GGCX:1502 KDM5A:2026
Macromolecule Biosynthetic Process (GO:0	−0.08395076	104	3.178e−03	4.917e−01	TNIP1:139 RPL28:224 RPLP0:234 MRPL27:352 SARS1:450 RPS5:541
Regulation Of JNK Cascade (GO:0046328)	−0.08920613	93	3.018e−03	4.917e−01	MAP3K11:80 IGF1R:101 CYLD:133 WNT7B:470 TRAF6:497 HMGB1:660
Regulation Of Type B Pancreatic Cell Pro	−0.38375568	5	2.962e−03	4.917e−01	SGPP2:677 PTPRN:1221 NR4A1:1516 NR4A3:1537 NUPR1:2446 NA
Negative Regulation Of SMAD Protein Sign	0.37634528	5	3.565e−03	5.039e−01	SLC2A10:50 PBLD:158 VEPH1:894 LRP1:1341 CILP:5418 NA
Acid Secretion (GO:0046717)	0.31623173	7	3.766e−03	5.057e−01	SLC22A16:24 DRD3:837 SLC51B:931 SLC51A:939 GHRL:1547 SLC26A6:4198
Positive Regulation Of Isotype Switching	0.23272398	13	3.676e−03	5.057e−01	HMCES:63 TGFB1:269 PAXIP1:902 KMT5C:924 KMT5B:1369 SHLD2:1655

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
BLALOCK_ALZHEIMERS_DISEASE_DN	−0.06204481	766	8.308e−09	5.380e−05	PTGES3:10 CDK17:15 UBL3:23 PLK2:54 CSNK1G3:60 ATP5IF1:81
REACTOME_REGULATION_OF_EXPRESSION_OF_SLI	−0.16876670	80	1.868e−07	6.050e−04	PSMA7:138 RNPS1:142 USP33:180 PSMD7:184 RPL28:224 RPLP0:234
REACTOME_THE_ROLE_OF_GTSE1_IN_G2_M_PROGR	−0.21293218	46	5.935e−07	1.281e−03	PSMA7:138 PSMD7:184 TP53:297 UBB:399 SEM1:702 PSMF1:809
HOUNKPE_HOUSEKEEPING_GENES	−0.05424055	712	1.129e−06	1.828e−03	SPAG9:18 TBL1XR1:40 COX15:42 BAP1:45 AP3M1:53 TMEM219:59
REACTOME_SIGNALING_BY_WNT	−0.11009639	162	1.419e−06	1.838e−03	PLCB3:51 KMT2D:75 PSMA7:138 VPS35:144 PRICKLE1:163 PRKG1:167
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	−0.12834661	113	2.545e−06	2.354e−03	PSMA7:138 RNPS1:142 USP33:180 PSMD7:184 PFN1:202 RPL28:224
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	−0.12112995	128	2.328e−06	2.354e−03	PHC1:73 KMT2D:75 PSMA7:138 PSMD7:184 CBX6:223 UBB:399
REACTOME_DISEASES_OF_SIGNAL_TRANSDUCTION	−0.07854398	297	3.613e−06	2.925e−03	RAP1A:1 POLR2E:21 TBL1XR1:40 MAP3K11:80 PSMA7:138 PSMD7:184
REACTOME_DEGRADATION_OF_GLI1_BY_THE_PROT	−0.23125302	32	6.023e−06	4.334e−03	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 NUMB:738 PSMF1:809
REACTOME_REGULATION_OF_MRNA_STABILITY_BY	−0.18948137	46	8.856e−06	5.735e−03	PSMA7:138 PSMD7:184 PABPC1:374 UBB:399 SEM1:702 PSMF1:809
MILI_PSEUDOPODIA_HAPTOTAXIS_UP	−0.07077197	328	1.182e−05	6.958e−03	RAP1A:1 AP3M1:53 CSNK1G3:60 PLEKHA8:84 ZRANB2:87 EIF2A:116
REACTOME_DEVELOPMENTAL_BIOLOGY	−0.04894056	681	1.700e−05	9.172e−03	NR5A2:12 SPAG9:18 POLR2E:21 EFNA5:29 TRPC5:31 GDNF:36
REACTOME_RUNX1_REGULATES_TRANSCRIPTION_O	−0.18262151	46	1.848e−05	9.206e−03	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_NEGATIVE_REGULATION_OF_NOTCH4_S	−0.23034815	27	3.449e−05	1.507e−02	PSMA7:138 PSMD7:184 UBB:399 TACC3:605 SEM1:702 PSMF1:809
CAIRO_HEPATOBLASTOMA_DN	0.08602525	196	3.491e−05	1.507e−02	CPT2:59 TMEM131L:77 SLC1A1:84 ABCB11:144 ACAA2:239 ACADM:307
REACTOME_TCF_DEPENDENT_SIGNALING_IN_RESP	−0.11105458	115	4.022e−05	1.557e−02	KMT2D:75 PSMA7:138 PSMD7:184 KREMEN2:381 UBB:399 SEM1:702
REACTOME_ASYMMETRIC_LOCALIZATION_OF_PCP_	−0.19712965	36	4.292e−05	1.557e−02	PSMA7:138 PRICKLE1:163 PSMD7:184 UBB:399 PARD6A:642 SEM1:702
REACTOME_DEFECTIVE_CFTR_CAUSES_CYSTIC_FI	−0.20577752	33	4.326e−05	1.557e−02	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_MAPK_FAMILY_SIGNALING_CASCADES	−0.07968579	219	5.150e−05	1.755e−02	RAP1A:1 GDNF:36 MAP3K11:80 IL5RA:88 PSMA7:138 PSMD7:184
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	−0.14034637	69	5.647e−05	1.829e−02	ITGA5:37 SMAD6:130 PSMA7:138 PSMD7:184 UBB:399 SEM1:702
REACTOME_REGULATION_OF_RAS_BY_GAPS	−0.18789741	38	6.173e−05	1.904e−02	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_AUF1_HNRNP_D0_BINDS_AND_DESTABI	−0.20721344	31	6.567e−05	1.933e−02	PSMA7:138 PSMD7:184 PABPC1:374 UBB:399 SEM1:702 PSMF1:809
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.07801675	219	7.373e−05	2.076e−02	SLC5A12:15 SLC22A16:24 SLC2A10:50 SLC1A1:84 RSC1A1:142 SLC36A2:227
HSIAO_HOUSEKEEPING_GENES	−0.08033602	204	8.084e−05	2.181e−02	ABLIM1:62 CD34:64 AARS1:170 PSMD7:184 PFN1:202 RPL28:224
REACTOME_SIGNALING_BY_THE_B_CELL_RECEPTO	−0.13658717	69	8.884e−05	2.301e−02	PSMA7:138 PSMD7:184 UBB:399 NFATC2:457 SEM1:702 MALT1:728
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	−0.05273123	470	1.026e−04	2.447e−02	RAP1A:1 ASB15:13 CD34:64 CD4:70 FBXW9:137 PSMA7:138
WP_NRF2_PATHWAY	0.11711941	92	1.058e−04	2.447e−02	SLC5A12:15 SLC2A10:50 TXNRD1:237 SLC2A11:264 TGFB1:269 SLC2A7:315
KEGG_LYSINE_DEGRADATION	0.18256884	38	9.922e−05	2.447e−02	NSD2:72 EHHADH:116 EHMT1:283 NSD3:352 GCDH:418 SETD1B:859
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.04803619	566	1.100e−04	2.456e−02	TRPM6:6 TRPM5:12 SLC5A12:15 SLC22A16:24 ADCY1:27 ATP9B:38
REACTOME_TRANSPORT_OF_BILE_SALTS_AND_ORG	0.12714593	77	1.168e−04	2.521e−02	SLC22A16:24 RSC1A1:142 SLC14A2:332 SLC13A2:389 SLC44A5:401 SLC6A6:519
REACTOME_STABILIZATION_OF_P53	−0.18888309	34	1.390e−04	2.905e−02	PSMA7:138 PSMD7:184 TP53:297 MDM2:317 UBB:399 SEM1:702
KIM_ALL_DISORDERS_CALB1_CORR_UP	−0.06009307	337	1.638e−04	3.315e−02	POLR2E:21 UBL3:23 BAP1:45 PLK2:54 RTN4:132 ATP6V1D:152
REACTOME_SIGNALING_BY_NOTCH4	−0.15317167	49	2.099e−04	4.120e−02	PSMA7:138 PSMD7:184 UBB:399 TACC3:605 MAML3:614 SEM1:702
REACTOME_NEDDYLATION	−0.08882418	146	2.191e−04	4.174e−02	ASB15:13 FBXW9:137 PSMA7:138 PSMD7:184 SPSB1:191 COPS6:346
REACTOME_DOWNSTREAM_SIGNALING_EVENTS_OF_	−0.15888063	45	2.285e−04	4.228e−02	PSMA7:138 PSMD7:184 UBB:399 NFATC2:457 SEM1:702 MALT1:728
REACTOME_UB_SPECIFIC_PROCESSING_PROTEASE	−0.10031610	111	2.677e−04	4.686e−02	USP19:103 CYLD:133 PSMA7:138 USP33:180 PSMD7:184 WDR20:240
REACTOME_REGULATION_OF_RUNX2_EXPRESSION_	−0.16886861	39	2.650e−04	4.686e−02	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_PCP_CE_PATHWAY	−0.14570295	52	2.814e−04	4.795e−02	PSMA7:138 PRICKLE1:163 PSMD7:184 PFN1:202 CLTC:348 UBB:399
ACEVEDO_LIVER_TUMOR_VS_NORMAL_ADJACENT_T	−0.04447811	579	2.947e−04	4.893e−02	AIF1L:90 PKP2:97 PPP1CA:104 TBL2:109 EIF2A:116 PRRC2C:117
REACTOME_SCF_BETA_TRCP_MEDIATED_DEGRADAT	−0.18959056	30	3.272e−04	5.105e−02	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Cholestasis	0.08695886	229	6.578e−06	6.426e−02	AFP:14 CLU:42 SLC23A1:76 ABCB11:144 KANK1:213 HSD17B4:385
Metabolic Bone Disorder	0.20114153	29	1.792e−04	5.833e−01	CYP2R1:92 ANKRD11:136 MMP2:514 PTH:520 CA2:701 GLB1:833
Nephrocalcinosis	0.16215230	45	1.702e−04	5.833e−01	TRPM6:6 CYP2R1:92 MRGPRF:147 CLDN16:186 ATP6V0A4:312 CDC73:470
Anemia, hereditary spherocytic hemolytic	0.41057786	5	1.476e−03	6.763e−01	SPTA1:123 ANK1:256 SLC4A1:670 EPB42:1416 SPTB:3113 NA
Cholestasis, chronic	0.36238981	5	5.012e−03	6.763e−01	ABCB11:144 AKR1D1:628 ABCB4:716 ATP8B1:1609 F2R:5529 NA
Cockayne Syndrome, Type II	−0.20418954	15	6.194e−03	6.763e−01	TP53:297 ERCC8:485 RIMS2:936 OGG1:938 IGF1:979 NELFE:1130
Inclusion Body Myopathy, Sporadic	−0.11246798	47	7.706e−03	6.763e−01	KCNH8:166 KCNH4:215 NT5C1A:295 UBB:399 TIMP1:464 OPTN:900
Lipomatosis, Multiple	−0.17784074	24	2.575e−03	6.763e−01	GDNF:36 BAP1:45 CD34:64 STAT5A:284 TP53:297 MDM2:317
Liposarcoma, Dedifferentiated	−0.16372566	27	3.250e−03	6.763e−01	CD34:64 IGF1R:101 TP53:297 MDM2:317 HOXA5:355 RB1:885
Polyarticular Juvenile Idiopathic Arthri	−0.18789728	17	7.333e−03	6.763e−01	LNPEP:1401 IL2RA:1451 LTBR:1630 CCR3:1790 STAT4:1869 STAT1:2047
Psychoses, Drug	0.23994118	15	1.297e−03	6.763e−01	CAMK2B:106 RGS9:183 HPGDS:878 CALR:966 DTNBP1:1013 HTR6:1628
Thymoma, type C	−0.20718557	14	7.286e−03	6.763e−01	IGF1R:101 TP53:297 HMGB1:660 KIT:807 ERBB2:1571 CD70:1763
22q11 Deletion Syndrome	0.22230733	16	2.085e−03	6.763e−01	HIRA:13 ZDHHC8:48 COMT:391 ARVCF:805 CHRD:948 FGF8:2009
Acute kidney injury	0.10032471	62	6.371e−03	6.763e−01	CLU:42 ALB:79 TGFB1:269 A2M:454 DGKE:498 AMN:522
Adenocarcinoma of rectum	0.15880843	24	7.107e−03	6.763e−01	ENTPD1:46 KRT20:486 NT5E:871 MYD88:1141 FOXP3:1202 DCC:1510
Anxiety state	0.24859235	11	4.311e−03	6.763e−01	IL1R2:98 COMT:391 TNF:1025 IL1A:1802 BDNF:2052 GDE1:2140
Atherogenesis	0.13230981	39	4.281e−03	6.763e−01	TRPV1:295 NPC1:561 HP:777 ABCG5:834 CNR2:945 TNF:1025
Benign melanocytic nevus	−0.11659454	48	5.246e−03	6.763e−01	BAP1:45 TP53:297 MDM2:317 CDK6:468 KIT:807 TUFM:869
Benign Neoplasm	0.12366498	47	3.390e−03	6.763e−01	KCNH2:133 BRCA1:243 TRPV1:295 GPR55:567 ATM:582 VDR:1208
Benign recurrent intrahepatic cholestasi	0.34796095	6	3.162e−03	6.763e−01	ABCB11:144 GGTLC3:406 ATP8B1:1609 GGT2:2467 GGT1:3285 GGTLC1:3482
Bilirubin measurement	0.22390196	15	2.686e−03	6.763e−01	PLA2R1:290 SLCO1A2:474 DGKD:719 MROH2A:734 SLCO1B1:860 DCHS2:1352
Broad chest	−0.32203611	6	6.302e−03	6.763e−01	XYLT2:342 CHST3:1288 MAP2K1:1722 TRAPPC2:1800 DYM:2725 PTPN11:5795
C−reactive protein measurement	0.14436647	33	4.130e−03	6.763e−01	NCAN:118 RORA:217 SUGP1:242 HPN:402 ABCG5:834 DOCK4:896
Calcium oxalate kidney stones	0.25709278	9	7.574e−03	6.763e−01	SLC36A2:227 AGXT:508 GRHPR:1735 SLC6A20:2794 OPLAH:2795 HOGA1:2989
Central neuroblastoma	−0.02894324	1086	1.788e−03	6.763e−01	RAP1A:1 NPFF:7 EDNRB:22 SMUG1:24 GDNF:36 ITGA5:37
Cerebral arterial aneurysm	0.10744103	57	5.079e−03	6.763e−01	TGFB1:269 ENG:467 MMP2:514 COL1A2:540 SCN11A:547 TMEM132B:891
Childhood Ependymoma	−0.24420382	10	7.502e−03	6.763e−01	TNC:63 TP53:297 ERBB4:453 PIK3CD:1596 PIK3CG:2551 PIK3CA:3086
Chronic kidney disease stage 5	0.04750796	275	7.102e−03	6.763e−01	ENTPD1:46 CPT2:59 ALB:79 CHDH:89 CUBN:120 TGFB1:269
Chronic Liver Failure	0.27018128	10	3.095e−03	6.763e−01	ABCB11:144 AKR1D1:628 IGFBP1:647 ABCB4:716 THPO:1404 EIF2AK3:2445
Ciliopathies	0.07255638	150	2.247e−03	6.763e−01	DYNC2H1:19 KIAA0753:176 MICAL3:191 TGFB1:269 TCTN1:280 ARL13B:284
Classical Hodgkin's Lymphoma	−0.08887957	116	9.772e−04	6.763e−01	IGF1R:101 CYLD:133 STAT5A:284 TP53:297 MDM2:317 IL10:428
Cold intolerance	0.33845953	6	4.093e−03	6.763e−01	TRPV1:295 LPIN1:986 UCP1:1123 TRPA1:1170 PPARA:1307 IRF4:7311
Common wart	−0.35596474	5	5.843e−03	6.763e−01	TP53:297 MDM2:317 HSPA4:1227 PCNA:2116 CDKN1A:5264 NA
Cone dysfunction syndrome	0.20705253	14	7.324e−03	6.763e−01	PROM1:80 RGS9:183 RGS9BP:574 PITPNM3:633 G6PD:1086 ADGRV1:2378
Congenital exomphalos	0.31345163	7	4.083e−03	6.763e−01	AFP:14 LRP1:1341 CHRNA7:1588 SDS:1829 SLC32A1:3565 GDF11:3854
Congenital hemivertebra	−0.16190383	25	5.099e−03	6.763e−01	RAP1A:1 KMT2D:75 TMCO1:248 JAG1:960 ATRX:1149 TBX6:1474
CREST Syndrome	−0.20310734	21	1.279e−03	6.763e−01	CCR1:241 IL10:428 TIMP1:464 HMGB1:660 PAH:813 CCR6:850
Cystic Fibrosis	0.05044350	374	9.108e−04	6.763e−01	BPIFB1:1 MUC20:4 AFP:14 CTRC:22 CELA1:41 ENTPD1:46
Cystic Kidney Diseases	0.10695454	61	3.915e−03	6.763e−01	TCTN1:280 ARL13B:284 TCTN3:528 TSC1:929 CEP164:1018 NPHP3:1082
Decreased muscle mass	0.15792537	28	3.844e−03	6.763e−01	NSD2:72 HERC2:145 ATP6V0A2:150 SPG11:286 LETM1:383 HSD17B4:385

KeggReactome Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.06089826	623	3.811e−07	6.871e−04	SLC5A12:4 TRPM6:5 CLCNKA:14 SLC14A2:17 CAMK2B:21 ADCY1:38
REACTOME_PEPTIDE_LIGAND_BINDING_RECEPTOR	−0.10230307	166	6.095e−06	5.494e−03	TACR3:39 GRP:50 NPFF:65 F2:170 APP:184 NMUR1:217
REACTOME_GLUTATHIONE_CONJUGATION	0.23843873	25	3.726e−05	1.933e−02	MGST3:74 GSTO1:338 CHAC2:355 GSTK1:869 MGST1:871 GSTZ1:1364
REACTOME_ION_CHANNEL_TRANSPORT	0.09253884	166	4.288e−05	1.933e−02	TRPM6:5 CLCNKA:14 CAMK2B:21 TRPM5:50 TRPV1:95 TRPM1:96
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.44967496	6	1.367e−04	4.107e−02	SLC4A1:156 RHAG:257 CA4:408 CA2:608 AQP1:1236 CA1:1740
REACTOME_PROCESSING_OF_CAPPED_INTRONLESS	0.25565218	19	1.153e−04	4.107e−02	SNRPD3:239 SLBP:383 NUDT21:702 SYMPK:741 CPSF7:1206 SNRPG:1830
REACTOME_MUSCLE_CONTRACTION	0.08240909	176	1.772e−04	4.564e−02	CAMK2B:21 KCNK15:71 ATP2B4:238 KCNK16:296 ACTC1:332 MYL4:450
REACTOME_SPERM_MOTILITY_AND_TAXES	0.36931054	8	2.984e−04	6.726e−02	CATSPERB:277 CATSPERG:820 CATSPER4:882 CATSPER2:1397 CATSPER3:1498 KCNU1:1620
REACTOME_STIMULI_SENSING_CHANNELS	0.10268506	101	3.789e−04	7.590e−02	TRPM6:5 CLCNKA:14 TRPM5:50 TRPV1:95 TRPM1:96 CLCNKB:140
REACTOME_RECYCLING_OF_BILE_ACIDS_AND_SAL	0.23588981	16	1.093e−03	7.612e−02	ALB:47 SLCO1A2:237 SLC10A2:260 ABCB11:262 SLCO1B1:558 SLC51A:924
REACTOME_GENERATION_OF_SECOND_MESSENGER_	−0.19365382	25	8.102e−04	7.612e−02	EVL:41 ZAP70:83 CD101:782 PAK2:802 CD247:807 HLA−DPA1:812
REACTOME_GENE_SILENCING_BY_RNA	0.11167511	82	4.889e−04	7.612e−02	MOV10L1:46 H3−3A:194 POLR2J:289 PIWIL2:322 RAN:486 PIWIL1:536
REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPT	−0.06029674	294	4.287e−04	7.612e−02	TACR3:39 GRP:50 NPFF:65 F2:170 APP:184 NMUR1:217
REACTOME_CLASS_C_3_METABOTROPIC_GLUTAMAT	0.18328153	27	9.881e−04	7.612e−02	TAS2R10:191 TAS2R41:862 GRM6:1111 TAS2R60:2042 TAS1R1:2108 TAS2R42:2250
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.09425009	101	1.105e−03	7.612e−02	SLC5A12:4 SLC6A6:142 SLC4A1:156 SLC1A1:320 SLC6A12:353 SLC26A3:412
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.06569806	232	6.240e−04	7.612e−02	SLC5A12:4 SLC14A2:17 SLC2A7:92 SLC6A6:142 SLC4A1:156 SLCO1A2:237
REACTOME_DNA_METHYLATION	0.17996388	28	9.897e−04	7.612e−02	H3−3A:194 DNMT1:354 DNMT3B:472 H4C15:637 H4C14:784 H2BC21:848
REACTOME_METABOLISM_OF_LIPIDS	0.03937332	650	8.134e−04	7.612e−02	EHHADH:32 ACADM:41 ALB:47 CYP1A1:82 PHYH:94 CYP17A1:105
REACTOME_PROCESSING_OF_DNA_DOUBLE_STRAND	0.11358383	69	1.132e−03	7.612e−02	PPP4C:139 HERC2:145 UBE2I:215 CLSPN:226 BRCA1:236 RNF168:251
REACTOME_CHOLINE_CATABOLISM	0.38561982	6	1.072e−03	7.612e−02	CHDH:391 DMGDH:526 ALDH7A1:1445 SARDH:1592 BHMT:2136 SLC44A1:3971
REACTOME_RNA_POLYMERASE_III_TRANSCRIPTIO	−0.24217347	16	8.010e−04	7.612e−02	POLR3G:958 POLR3B:1098 NFIA:1750 POLR3GL:1843 CRCP:1974 SSB:2497
REACTOME_METABOLISM_OF_STEROIDS	0.08529531	132	7.535e−04	7.612e−02	ALB:47 CYP17A1:105 CUBN:124 UBE2I:215 SLCO1A2:237 CYP11A1:243
REACTOME_TRIGLYCERIDE_METABOLISM	0.16170518	34	1.115e−03	7.612e−02	GPAT2:107 FABP1:213 MOGAT1:359 LPIN1:366 LPIN2:599 MOGAT2:1163
REACTOME_ION_TRANSPORT_BY_P_TYPE_ATPASES	0.13692990	47	1.182e−03	7.612e−02	CAMK2B:21 ATP2B4:238 ATP9B:427 ATP2A1:566 ATP8B1:645 CAMK2D:682
REACTOME_SENSORY_PERCEPTION_OF_TASTE	0.16893705	35	5.499e−04	7.612e−02	TRPM5:50 TAS2R10:191 TAS2R41:862 SCNN1B:1050 CALHM1:1158 SCNN1A:1240
REACTOME_SARS_COV_2_MODULATES_HOST_TRANS	0.19482657	26	5.901e−04	7.612e−02	SNRPD1:130 SNRPD3:239 RPS14:249 RPS27A:317 RPS15A:335 RPS23:521
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.17741974	28	1.167e−03	7.612e−02	CYP1A1:82 CYP17A1:105 CYP11A1:243 COMT:280 HSD11B1:561 SULT1E1:701
KEGG_STARCH_AND_SUCROSE_METABOLISM	0.18052568	28	9.541e−04	7.612e−02	SI:304 ENPP3:508 HK2:625 GYS2:1035 GUSB:1041 UGDH:1581
REACTOME_STRIATED_MUSCLE_CONTRACTION	0.16641166	31	1.356e−03	8.430e−02	ACTC1:332 MYL4:450 DES:534 NEB:947 TNNI2:961 MYBPC2:1145
REACTOME_COMPLEX_I_BIOGENESIS	0.13045937	48	1.794e−03	9.929e−02	NDUFB8:53 NDUFS5:123 NDUFS3:220 NDUFA9:295 ECSIT:549 NDUFB5:611
REACTOME_PURINE_SALVAGE	−0.25054144	13	1.767e−03	9.929e−02	AMPD1:112 GMPR2:619 DGUOK:697 APRT:742 PNP:942 ADAL:1434
REACTOME_SLBP_DEPENDENT_PROCESSING_OF_RE	0.36933049	6	1.732e−03	9.929e−02	SNRPD3:239 SLBP:383 SNRPG:1830 NCBP1:2395 LSM11:2422 LSM10:4285
REACTOME_PARACETAMOL_ADME	0.21856331	17	1.817e−03	9.929e−02	ACY1:60 SULT1A3:148 SULT1A4:154 SULT1E1:701 ABCC5:898 GGT5:1604
REACTOME_RUNX1_REGULATES_GENES_INVOLVED_	0.12631223	50	2.034e−03	1.079e−01	H3−3A:194 H4C15:637 H4C14:784 H2BC21:848 H4C2:881 MOV10:1422
REACTOME_GLUCOCORTICOID_BIOSYNTHESIS	0.35791676	6	2.399e−03	1.211e−01	CYP17A1:105 HSD11B1:561 CYP21A2:1294 SERPINA6:1769 HSD3B2:3480 HSD11B2:5338
REACTOME_G_ALPHA_Q_SIGNALLING_EVENTS	−0.06476625	185	2.531e−03	1.211e−01	TACR3:39 GRP:50 NPFF:65 DGKH:80 F2:170 RGS16:174
REACTOME_INTERACTION_BETWEEN_L1_AND_ANKY	0.15887465	30	2.619e−03	1.211e−01	ANK1:66 SPTA1:84 KCNQ3:202 L1CAM:318 SCN11A:710 SPTBN5:759
REACTOME_PIWI_INTERACTING_RNA_PIRNA_BIOG	0.21177725	17	2.513e−03	1.211e−01	MOV10L1:46 POLR2J:289 PIWIL2:322 PIWIL1:536 ASZ1:1675 TDRD12:2329
KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRA	0.13298216	43	2.584e−03	1.211e−01	EHHADH:32 ACADM:41 ACAD8:443 BCAT2:585 HIBCH:658 ACAA2:1060
REACTOME_PREDNISONE_ADME	0.38591281	5	2.806e−03	1.265e−01	ALB:47 HSD11B1:561 ABCB1:676 SERPINA6:1769 HSD11B2:5338 NA

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0001666 abnormal nutrient absorption	0.19618624	40	1.888e−05	8.986e−03	CLPS:45 SLC23A1:76 ALB:79 PNLIP:129 RSC1A1:142 HSD17B4:385
MP0002118 abnormal lipid homeostasis	0.06531422	338	6.527e−05	1.553e−02	INPP5K:35 CLPS:45 ALB:79 CYP11A1:82 EHHADH:116 PNLIP:129
MP0000609 abnormal liver physiology	0.05755144	322	5.797e−04	7.402e−02	CRTC2:11 MAN2A1:34 CYBRD1:66 ALB:79 EHHADH:116 ABCB11:144
MP0002136 abnormal kidney physiology	0.06426941	245	7.144e−04	7.402e−02	MAN2A1:34 SLC23A1:76 ALB:79 CUBN:120 CLDN16:186 AQP11:196
MP0001929 abnormal gametogenesis	0.05321350	363	7.775e−04	7.402e−02	MOV10L1:51 CYP11A1:82 FANCM:121 SMPD3:128 PADI6:137 HERC2:145
MP0003868 abnormal feces composition	0.18227467	24	2.040e−03	1.337e−01	CLPS:45 ALB:79 PNLIP:129 ABCB4:716 CYP7A1:882 FOXP3:1202
MP0009643 abnormal urine homeostasis	0.06313199	200	2.528e−03	1.337e−01	MAN2A1:34 SLC23A1:76 ALB:79 SLC1A1:84 CUBN:120 CLDN16:186
MP0004019 abnormal vitamin homeostasis	0.15272128	34	2.124e−03	1.337e−01	SLC23A1:76 CUBN:120 CLDN16:186 RBP2:450 TRPV5:578 LRAT:678
MP0002184 abnormal innervation	−0.08732870	104	2.309e−03	1.337e−01	EFNA5:29 GDNF:36 CHRNA9:243 ERBB4:453 NFATC2:457 RIMS1:481
MP0000955 abnormal spinal cord	−0.06536695	161	4.822e−03	1.957e−01	AMBRA1:34 GDNF:36 VSX2:44 IGF1R:101 SLC1A2:190 ATCAY:325
MP0005085 abnormal gallbladder physiolo	0.16969289	23	4.934e−03	1.957e−01	ABCB11:144 ABCB4:716 ABCG5:834 CYP7A1:882 F2RL1:919 GPBAR1:1031
MP0001756 abnormal urination	0.09805776	70	4.825e−03	1.957e−01	ALB:79 DST:112 CHRNB2:139 TRPV1:295 SLC14A2:332 BCAT2:350
MP0002210 abnormal sex determination	0.05082020	263	5.670e−03	2.076e−01	MOV10L1:51 CYP11A1:82 FANCM:121 SMPD3:128 GNRH1:174 HIP1:224
MP0003252 abnormal bile duct	0.17673498	20	6.309e−03	2.145e−01	ABCB11:144 ABCB4:716 ABCG5:834 PKHD1:1102 ATP8B1:1609 PKD2:2263
MP0008875 abnormal xenobiotic pharmacok	0.21577272	13	7.132e−03	2.263e−01	CYP1A1:318 COMT:391 MTF1:504 SLC22A1:690 ABCC3:2112 SLC47A1:2169
MP0002078 abnormal glucose homeostasis	0.04119747	368	8.844e−03	2.355e−01	CRTC2:11 AFP:14 INPP5K:35 PNLIP:129 ABCB11:144 ENTPD5:258
MP0005448 abnormal energy balance	0.08001056	91	8.906e−03	2.355e−01	INPP5K:35 PNLIP:129 TRPV1:295 RPTOR:341 BCAT2:350 CACNA1B:472
MP0005395 other phenotype	0.09273826	69	8.126e−03	2.355e−01	ENTPD1:46 SLC23A1:76 DST:112 RBP2:450 CPT1A:502 AGXT:508
MP0001188 hyperpigmentation	−0.26529842	8	9.413e−03	2.358e−01	KIT:807 TERF2IP:834 TERF1:1476 GNAQ:2117 MC1R:2201 GNA11:3659
MP0000598 abnormal liver morphology	0.03821029	391	1.257e−02	2.610e−01	MAN2A1:34 NAGLU:44 CYBRD1:66 PNLIP:129 ABCB11:144 GNRH1:174
MP0000685 abnormal immune system	−0.12050023	36	1.264e−02	2.610e−01	EBI3:48 TNFSF11:669 MALT1:728 LTBR:1630 AIRE:2011 FOXN1:2130
MP0009642 abnormal blood homeostasis	0.02553044	1034	1.316e−02	2.610e−01	TRPM6:6 CRTC2:11 AFP:14 TRAF2:21 INPP5K:35 CLPS:45
MP0005319 abnormal enzyme/ coenzyme	0.06097622	144	1.271e−02	2.610e−01	SLC1A1:84 ENTPD5:258 TGFB1:269 UPP1:305 CYP1A1:318 HPN:402
MP0005410 abnormal fertilization	0.09914184	53	1.297e−02	2.610e−01	ADAM2:135 ACRBP:149 RIMBP3:304 ROS1:382 DKKL1:444 NPC1:561
MP0002135 abnormal kidney morphology	0.03817729	380	1.383e−02	2.634e−01	MAN2A1:34 NAGLU:44 SLC2A10:50 CUBN:120 GNRH1:174 MANBA:189
MP0005197 abnormal uvea morphology	−0.08638055	67	1.510e−02	2.700e−01	ALKBH1:194 TYRP1:296 IL10:428 CYP1B1:590 EFEMP1:593 HPS1:682
MP0005174 abnormal tail pigmentation	−0.18107312	15	1.532e−02	2.700e−01	EDNRB:22 HPS1:682 KIT:807 MITF:2039 GNAQ:2117 MC1R:2201
MP0000538 abnormal urinary bladder	0.11863661	34	1.700e−02	2.891e−01	NAGLU:44 CHRNB2:139 ITGB4:605 MEP1A:741 CHRNA3:843 ANXA4:883
MP0001765 abnormal ion homeostasis	0.07106152	92	1.952e−02	3.096e−01	TRPM6:6 CYP11A1:82 CLDN16:186 TRPV5:578 SLC12A3:626 SLC4A1:670
MP0002877 abnormal melanocyte morpholog	−0.23914306	8	1.925e−02	3.096e−01	KIT:807 RB1:885 ECE1:1000 LMNA:1660 MITF:2039 CDKN2A:5035
MP0002177 abnormal outer ear	−0.08429949	63	2.141e−02	3.288e−01	HIC1:225 HOXA5:355 HMX1:429 DDR1:620 HPS1:682 KIT:807
MP0005636 abnormal mineral homeostasis	0.06122255	118	2.310e−02	3.308e−01	TRPM6:6 CYBRD1:66 ALB:79 CYP11A1:82 CLDN16:186 ANK1:256
MP0001346 abnormal lacrimal gland	−0.20645911	10	2.390e−02	3.308e−01	VSX2:44 FGFR2:955 CCR7:1798 PRLR:1834 AIRE:2011 FAS:2964
MP0002139 abnormal hepatobiliary system	0.24493350	7	2.491e−02	3.308e−01	ALDH1A1:114 SLC22A1:690 ABCB4:716 FABP1:952 ABCG8:2890 INSIG1:7528
MP0005408 hypopigmentation	−0.15649085	17	2.572e−02	3.308e−01	PAH:813 ATP7B:1266 TYR:1268 MBTPS1:1851 MITF:2039 ERBB3:2486
MP0001186 pigmentation phenotype	−0.06699143	95	2.531e−02	3.308e−01	SPAG9:18 EDNRB:22 RABGGTA:32 TYRP1:296 HPS1:682 PAH:813
MP0010094 abnormal chromosome stability	0.08699686	58	2.264e−02	3.308e−01	MBD1:81 BRCA1:243 XRCC5:248 MCPH1:384 PMS2:414 ATM:582
MP0004811 abnormal neuron physiology	−0.04375600	222	2.786e−02	3.490e−01	EFNA5:29 ITGA5:37 CEND1:89 MFN2:92 SIM1:257 YWHAE:277
MP0003705 abnormal hypodermis morpholog	0.25402352	6	3.126e−02	3.816e−01	BRCA1:243 FBN1:1761 HR:2514 CBS:2645 SMAD4:4997 DSE:7039
MP0002734 abnormal mechanical nocicepti	0.08281714	55	3.454e−02	4.110e−01	ADCY1:27 DST:112 KIF1A:236 TRPV1:295 DBH:407 CACNA1B:472